

Enhancing USB Security: A Multi-Layered Framework for Detecting Vulnerabilities and Mitigating BadUSB Attacks

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B.Sc. in Computer Science and Engineering

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

USB devices, an indispensable object in modern computing with its increasing prevalence have introduced us to a significant cybersecurity risk. Studies say that 25% of malware infections originate from USB devices which makes them one of the most exploited attack vectors globally [35]. Among these, "bad USB" devices, such as reprogrammed USB controllers (e.g., BadUSB exploits) and hardware key-loggers pose severe threats by covertly capturing sensitive data, installing malware, or executing unauthorized commands. Alarmingly, global reports reveal a rising trend in USB-borne attacks with 9% of cybersecurity incidents linked to malicious USB devices [19]. This research proposes a comprehensive software framework solution to detect, mitigate threats, and neutralize malicious USB peripherals by developing a multi-layered security approach that combines USB sniffing, behavioral monitoring, anomaly detection, and layered authentication mechanisms. As key features it also includes USB metadata validation, CAPTCHA prompts to safeguard automated devices, and device fingerprinting to verify legitimacy without compromising usability. For example, while connecting a USB peripheral it will trigger and will pop up CAPTCHA verification alongside checking power consumption profiling and keystroke timing to identify potential threats. At the core of the framework, it includes machine learning, enabling dynamic adaptability and precision. By analyzing keystroke patterns, power consumption profiling, and USB activity logs, the system will differentiate malicious devices. While training on diverse datasets, it ensures that the models can detect evolving threats while no false positives, maintaining both security and user convenience. Testing results of this framework will validate its effectiveness in detecting and mitigating USB threats with high accuracy and no very low false positive rate. This practical solution fills a critical cybersecurity gap by safeguarding personal and enterprise systems from the growing menace of bad USB devices.

Keywords: BadUSB, USB threat mitigation, USB sniffing, Behavioral monitoring, Anomaly detection, Layered authentication, USB metadata validation, CAPTCHA verification, Device fingerprinting, Keystroke timing analysis, Power consumption profiling, dynamic threat detection, USB activity logs, false positive reduction.

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Chapter 1

Introduction

The digital era depends on USB devices to establish data transfers while creating peripheral connections that link hardware systems together. The extensive use of USB devices creates multiple security weaknesses that cyber attackers exploit to execute their primary attacks. Research shows that USB-borne malware attacks grew substantially from 2019 to 2024 with a 42% rise that brought the total to 51% [30]. Malicious USB devices can trick security systems by impersonating genuine peripherals to launch complex keylogging attacks, data theft operations, and firmware manipulation attacks [2].

Multiple high-profile cybersecurity organizations including the Cybersecurity and Infrastructure Security Agency (CISA) have warned about increasing USB exploitation risks. USB devices used for malicious purposes can automatically type keys while delivering payloads and taking control of system functions without evading antivirus software or endpoint protection. Attackers deploy pre-programmed microcontrollers including BadUSB, USB Rubber Ducky and Arduino-based payload injectors to perform stealthy intrusions which make detection more difficult [27].

Traditional USB security solutions primarily use software-based defenses to combat threats but they do not protect against hardware vulnerabilities at the low level. The security gap requires multi-layered protection systems that integrate anomaly detection with real-time monitoring and power consumption profiling behavioral analysis and CAPTCHA-based authentication [22]. Our research creates an advanced detection framework that detects malicious USB activities through hardware and software analysis and user interaction monitoring to provide complete protection.

The proposed framework uses machine learning algorithms to analyze human behavioral patterns through keystroke dynamics typing speed and USB interaction latency for attack detection. The framework implements USB metadata validation and power consumption profiling alongside AI-driven anomaly detection models to stop unauthorized USB activity without compromising user convenience or system functionality [16].

The growing reliance on USB peripherals throughout personal corporate and industrial environments makes USB communication security an essential requirement rather than an optional feature. The research actively identifies and mitigates mali-

cious USB threats to strengthen cybersecurity protocols and minimize unauthorized access risks while offering a scalable future-proof USB security model for individual users and enterprise-level infrastructure [10].

1.1 Background

Universal Serial Bus (USB) peripherals serve as fundamental components of contemporary computing systems because they facilitate effortless data transfer, device connectivity, and power delivery. The widespread adoption of USB peripherals throughout personal, professional, and industrial applications has simplified workflows and accelerated modern innovation. The widespread adoption of USB devices has created important security weaknesses that many users fail to recognize or often overlook [14].

The main security risk from USB peripherals emerges from their ability to be exploited by cyber attackers. Devices such as ATtiny85-based microcontrollers, and USB Rubber Ducky tools function as legitimate peripheral imitators that bypass standard security protocols. The inherent trust in USB technology allows these devices to execute payloads such as keyloggers data-exfiltration tools and sophisticated threats. The growing number of hidden attack vectors requires stronger protective measures for system security [41].

The rapid development of USB technology in hardware and firmware capabilities has outpaced the security development rate. Current security solutions primarily react to software vulnerabilities while ignoring threats that exist at hardware and physical levels. The lack of alignment between security development and hardware capabilities has resulted in substantial gaps within cybersecurity frameworks which expose systems to sophisticated threats that evade standard defense mechanisms [14].

Critical infrastructure sectors such as healthcare and finance together with industrial systems experience the most severe threats from malicious USB devices. Operations can break down while sensitive information leaks and financial and reputational damage becomes significant when a single USB device gets compromised. The destructive results of USB-borne attacks against industrial control systems and hospital networks demonstrate the necessity to develop proactive solutions [12].

Multiple security layers represent the necessary framework to reduce these threats. The combination of anomaly detection with real-time monitoring and adaptive authentication establishes a complete defense system against USB-based attacks. Early detection of malicious activity becomes possible through methods that combine device behavior anomaly detection with USB metadata validation and power consumption monitoring. These frameworks show the potential to transform USB security standards through their ability to maintain user convenience while achieving cross-environment compatibility.

The research fills essential gaps in USB security through a comprehensive framework that combines advanced methods including device fingerprinting behavioral analysis

and metadata validation. The proposed outcome creates an adaptable security standard for USB protection which defends personal users and essential systems against modern security threats.

1.2 Rational of the Study or Motivation

The rapid increase of USB peripherals in personal, professional, and organizational environments has introduced serious security threats. Though USB devices offer convenient data transfer, device connectivity, and power delivery, their inherent attributes turn them into dangerous weapons for malicious exploitation. This study is driven by the pressing need to address and mitigate critical vulnerabilities in cyber security caused by USB devices to ensure reliable and comfortable computing environments.

1.2.1 Limitations of the existing systems

Existing USB security systems mostly focus on software-level defenses and fail to address sophisticated hardware-level attacks. Malicious devices, such as USB Rubber Ducky, Arduino Pro Micro, ATtiny85 microcontrollers, and Hack5 tools, are capable of emulating legitimate peripherals to deploy covert payloads. These devices can go past traditional security systems and execute complex tasks like data exfiltration, keylogging, and injecting malicious commands [22]. Additionally, many current frameworks can analyze behavioral patterns or detect anomalies at the hardware and firmware levels, leaving critical vulnerabilities unaddressed and making USB devices a significant threat vector [22].

1.2.2 Relevance to Real-world circumstances

USB-based threats are particularly severe in critical infrastructure sectors such as finance, healthcare, and manufacturing. A malicious USB device has the potential to expose sensitive information, disrupt essential operations, and cause substantial financial and reputational loss. Attacks on industrial management systems could cause expensive operational disturbance [10] and malicious USB peripherals invading hospital networks could jeopardize patient data affecting the delivery of life-critical services. The above situations emphasize the importance of a dynamic and multi-layered defense strategy capable of correctly identifying and mitigating risks caused by USB peripherals.

1.2.3 Impact of the Study

The goal of this research is to reduce the existing gaps in USB security systems by developing a strong framework that integrates advanced techniques, including:

- **Behavioral Analysis:** Anomaly studies of device behavior help detect malicious intentions.
- **CAPTCHA-Based Validation:** A user-friendly approach of verification to restrict unauthorized access.

- **USB Metadata Validation:** Verify the metadata information of that USB device to measure its authenticity.

The proposed solution is designed to accomplish:

- Strong detection and prevention capabilities to detect malicious USB devices.
- The system reduces organizational disruptions and damage while maintaining user satisfaction.
- Protection of sensitive data and important systems from unauthorized access.

Our research addresses cybersecurity challenges through an adaptable framework that seeks to establish security across multiple digital environments.

1.3 Problem Statement

USB devices have now become a common feature in daily computer use, enabling people to transfer files, add peripherals and generally make things happen with simplicity. Such convenience, however, does not come without enormous security risks. Attackers have discovered methods of misusing the implicit trust that computers place in USB connections over the years. Most concerning of all is the rise of so-called BadUSB attacks in which a seemingly normal USB drive or keyboard is reprogrammed in a way that is undetectable to perform malicious tasks including keylogging, infecting a system with malware, or in some cases gaining complete control of a system.

The thing that is especially treacherous about these attacks is that they are stealthy. The majority of the common security controls, e.g., antivirus software or firewalls, are set up to identify threats on the software level. They will tend to miss the activity at the physical or behavioral layer, such as the minor variations in typing rates, unusual power consumption, or a device impersonating another device. Rogue USBs can therefore bypass security very easily and infect personal and organizational systems.

Although developers and researchers have devised measures to assist in USB security tightening, the majority of available solutions lack a comprehensive real time approach. The solution that is needed is one that is intelligent, multi-layered, and can not only detect the threat at an early stage but also one that can learn new forms of attacks and also be simple to use.

The study is aimed at bridging that gap. The key issue is that there is no decent system that would have capabilities to proactively analyze USB activity, learn normal patterns that indicate an attack, and defend users, without becoming obtrusive. Being capable of cracking it would result in a more secure computing world where USBs can be utilized without trepidation particularly in security-sensitive environments such as health care, finance, and key infrastructure.

1.4 Objective

The core goal of the proposed work is to design an effective and smart framework that can identify, prevent, and neutralize malicious USB devices, especially the ones that are using BadUSB-based attacks. Following the increasing risk of cyber-attacks through USB peripherals, the proposed research seeks to implement a multi-layered security mechanism consisting of real-time monitoring, behavioral analysis, USB metadata validation, and anomaly detection using machine learning.

Detect Malicious USB Devices:

- A machine learning detection system would be used to examine and analyze human behavioral data and USB device activity together with metadata in the way to detect risky USB devices.
- The system implements anomaly detection through methods which identify irregularities in USB activity patterns.

Mitigate USB Threats:

- Security protection functions through multiple layers combining human behavioral analysis with anomaly detection and threat identification to block USB-based attacks.
- Develop a threat mitigation system that stops recognized threats without interrupting regular user activities.

Enhance Usability and Accessibility:

- Users need access to a framework interface which enables simple system integration across multiple operational environments and devices.
- The users do not need to wait because the security is kept at a high level with the use of adaptive systems that connect CAPTCHA validation with device fingerprinting efforts.

Leverage Machine Learning for Anomaly Detection:

- Machine learning models need training and deployment to learn human-device interaction patterns before they can detect suspicious activity signals.
- Threat monitoring activities on collected insights support better detection mechanism development and threat adaptation.

Protect Data and Critical Systems:

- Protect sensitive data and maintain operational system integrity through protective measures in healthcare facilities and financial organizations and industrial production facilities.
- The security system should block only malicious USB devices yet allow legitimate ones without creating unnecessary blockages to protect users from false alerts.

At the end, the study provides a scalable, proactive defense mechanism that can be seamlessly deployed into personal or enterprise settings - protecting systems against a growing number of USB-based cyber attacks.

1.5 Methodology in Brief

To find out the vulnerabilities of USB peripherals and upgrade their security through advanced detection and prevention techniques, this study adopts a multi-faceted approach. Firstly to detect lackings in existing security measures at both hardware and software levels, this study proposes an in-depth analysis of USB-related risks, such as rogue peripherals and malicious device behaviors. Next, A dynamic and multi-layered security framework is developed containing various user authentication methods including behavioral analysis, CAPTCHA verification, USB metadata validation, power consumption profiling, and device fingerprinting. This framework aims to conduct dynamic identification and prevention of malicious USB activities. To accurately achieve that, we created datasets collecting human behavioral patterns such as keystroke dynamics and timings for both legitimate and malicious devices to validate and support the training and testing of our security framework. Then, The detection framework is developed in the implementation phase combining machine learning models and rule-based algorithms, incorporating features like keystroke latency analysis, anomaly detection, and power consumption profiling. The framework will undergo precise and continuous testing of multiple USB peripherals using known malicious tools to validate its detection accuracy, response times, and false positive rates across simulated and real-world environments. The framework evaluation process focuses on threat detection capabilities as well as usability and scalability improvements promoting legitimate USB functionality and user convenience relative to existing solutions. Our research documents all of our findings alongside implementation details and limitations while we concentrate on framework optimization through testing feedback and newly discovered constraints.

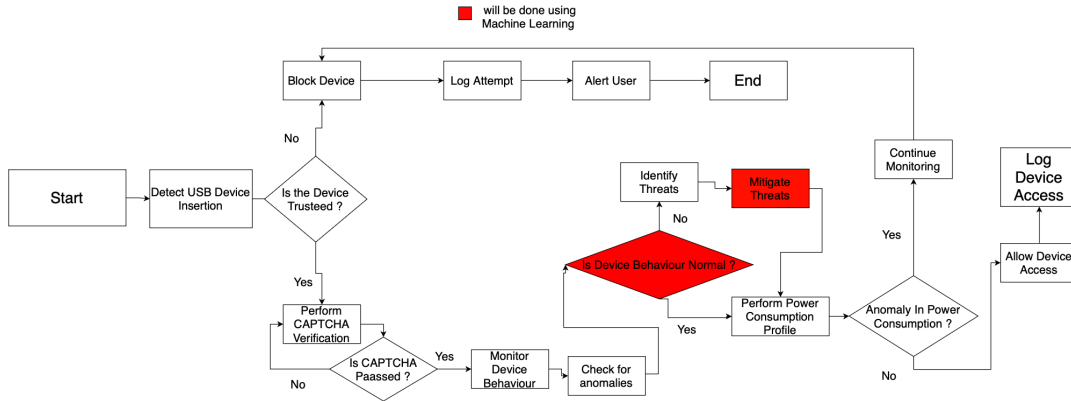


Figure 1.1: Methodology

1.6 Scopes and Challenges

1.6.1 Scopes of the Research

Proactive Threat Detection

In recent times, the identification of malicious USB devices has advanced detection

techniques that are developing, such as anomaly detection, behavioral analysis, and USB metadata validation.

Layered Security Measures

Deploying adaptive security elements that rely on CAPTCHA authentication and device fingerprinting security to block unwanted device access attempts.

Integration with Diverse Environments

Design the framework to operate well with personal computers as well as enterprise networks and critical infrastructure sectors that cover healthcare, finance, and manufacturing industries.

Real-Time Monitoring and Response

Develop real-time threat identification technology that promptly responds to dangerous occurrences while maintaining all authorized USB peripheral functionality.

Advancing Cybersecurity Practices

Keep contributing to developing USB security standards together with best practice recommendations that identify new attack approaches while maintaining user-focused designs.

1.6.2 Challenges of the Research

Data Collection and Diversity

This represents significant challenges for collecting comprehensive human behavioral data, like keystroke dynamics, typing patterns, and interaction timings. Capturing a variety of datasets that are required from users with typing styles and habits while simulating real-time scenarios to include accurate and malicious activities for precise testing and validation.

Balancing Security and Usability

Users need security measures that combine ease of collaboration with intuitive design. The system must prevent the generation of pointless alert notifications.

Scalability and Resource Efficiency

The framework design needs to prove its ability to scale and use resources efficiently when operating between different platforms including enterprise systems and IoT devices without creating excessive operational or computational demands.

Adapting to Emerging Threats

The framework must maintain adaptability while remaining resistant to new attack techniques that attackers develop against USB vulnerabilities. The fight against emerging security threats requires continuous development and system updates as the fundamental method for protection.

Chapter 2

Literature Review

2.1 Preliminaries

2.1.1 Background Concepts

The research demands knowledge about how USB devices operate alongside understanding malicious USB devices and security mechanisms to define its scope and research methodologies.

- **Universal Serial Bus (USB) Technology**

The Universal Serial Bus (USB) functions as the global standard for data transmission and device connection and power delivery. The plug-and-play functionality combined with versatility positions makes this device an essential component in contemporary computing environments [34]. The device's versatile features enable malicious actors to exploit it [32].

- **Malicious USB Devices**

The exploitation of USB protocols enables malicious devices to carry out their attacks. The interfaces serve as delivery channels for unauthorized payloads which include keyloggers, ransomware, and data exfiltration tools. The combination of USB Rubber Ducky and ATtiny85-based microcontrollers enables attackers to create legitimate peripheral emulations that bypass standard security protocols to threaten system security [40].

- **USB Threat Vectors**

USB-based threats manifest at multiple levels:

Hardware Layer: The use of compromised firmware or counterfeit devices tricks systems into believing they are working with authentic peripherals.

Software Layer: When a USB device connects to a system malicious scripts automatically activate to obtain unauthorized system access.

Behavioral Layer: Unusual device behavior detected through unexpected data movements and abnormal keystroke patterns indicates potential malicious design elements.

2.1.2 Definitions and Terminology

CAPTCHA Validation

The authentication tool CAPTCHA differentiates between human users and automated scripts. The security system based on USB verification protects important systems by preventing unauthorized peripheral devices from gaining access [3].

Device Fingerprinting

The method builds unique USB device identification through a combination of metadata and serial numbers and power consumption profiles. Through authentication processes, device fingerprinting strengthens device security by verifying device legitimacy.

Behavioral Analysis

The analysis of USB device patterns through systematic examination reveals abnormal behavioral deviations. Anomaly detection methods function as fundamental tools to detect malicious activities.

Power Consumption Profiling

The power usage of USB devices serves as a detection method for identifying abnormal activities. Research indicates that malicious devices create distinctive power usage signatures that help distinguish them from ordinary peripherals.

Real-Time Monitoring

The proactive security approach needs continuous USB activity monitoring to identify threats before they can be effectively responded to.

2.1.3 Relevant Theories and Models

Multi-Layered Security Models

Security frameworks protect systems with multiple detection and prevention layers that defend against vulnerabilities that propagate across various system levels. The principle protects systems against threats that propagate through USB devices.

Anomaly Detection in Cybersecurity

Machine learning-based anomaly detection systems detect suspicious USB behavior patterns that signal potential malicious activity according to research [31].

Principle of Least Privilege

Security protocols that limit USB device permissions to essential levels create minimal attack surfaces for malicious devices.

The foundation section introduces essential concepts and technical terms that are relevant to USB security. The inclusion of references builds credibility by linking research findings to established knowledge.

2.2 Review of Existing Research

Now that we have a clearly planned strategy, we studied papers and articles that closely relate to our research from reputed sources such as IEEE, ResearchGate, and ScienceDirect to gain more insights. These studies address USB device security, hardware and firmware vulnerabilities, anomaly detection, and machine learning applications in cybersecurity. We learned more about the risks posed by USB devices, the evolving threat landscape, and advanced detection mechanisms. By reviewing these works, we have gained a comprehensive understanding of the challenges and opportunities in developing a robust and scalable USB security framework. Some of our readings are overviewed here:

To begin with, This study [9] evaluates the vulnerabilities within USB peripherals by tracing their evolution from USB 1.0 to USB-C. The paper addresses design flaws including lacking in authentication techniques and encryption in earlier USB versions, which make the systems vulnerable to unauthorized data access, firmware exploitation, and malware injection. It also addresses the threats associated with the availability of USB devices, emphasizing the pressing need for secure boot mechanisms, upgraded firmware protection strategies, and metadata validation to eliminate these vulnerabilities.

Furthermore, An article [30] demonstrated that there has been an increase in malware attacks transmitted through USB devices across industrial facilities. Research shows USB devices have become responsible for 51% of malware attacks in 2024 while their share has more than doubled in just five years from the original 9% in 2019. The rising trend demonstrates how industrial environments increasingly use USB devices to initiate cyberattacks. The report shows that content-based malware has become more prevalent because it exploits existing documents and scripting functions to represent 20% of detected malware. The analysis revealed that 13% of blocked malware used Word, Excel, and PDF document formats to launch attacks. These findings demonstrate the urgency to deploy stronger cybersecurity standards by implementing strict USB device monitoring alongside state-of-the-art threat detection solutions together with public employee security training for keeping industrial facilities protected from USB-based dangers

Moreover, the sectors of manufacturing and transportation along with healthcare and finance use removable media primarily through USB drives to execute essential operations such as software maintenance and firmware updates in air-gapped operational technology (OT) systems. This paper [28] addresses how USB devices have evolved into attractive targets for cybercriminals because they bypass security protocols and allow the entry of malware despite the 2023 Sogu malware attack which targeted multinational companies. The advancing complexity of attacks now includes malicious USB-based keystroke injection that allows attackers to break into secure networks to steal sensitive information. Removable media security proves difficult to manage because organizations lack standard security rules and incomplete visibility into file transfer operations which exposes numerous businesses to potential threats. The prevention of these security threats requires organizations to develop extensive scanning protocols together with Content Disarm and Reconstruction (CDR) meth-

ods and enforce stringent authorization controls and data encryption systems. A combination of continuous security audits alongside systematic employee training and multiple security measures establishes absolute protection against malware intrusion while reducing vulnerability to security incidents.

Ranney et al. (2025) [39] introduces a surprisingly clever technique called USB-Snoop which is a side-channel attack, spying on user activity by taking advantage of how USB hubs manage traffic. The main motto of the method is that a seemingly harmless device like a regular USB device can be used to overload the USB connection with small bursts of data, causing delays. When these delays are precisely measured, it becomes possible to figure out what other devices are doing. For instance, when a user types on a keyboard, those keystrokes create specific timing patterns in data traffic which can be guessed by using machine learning techniques like Hidden Markov Models. Similarly, by watching the timing of network traffic from a USB Ethernet adapter, the system can monitor which websites the user is visiting- achieving the degree of accuracy over 83% using deep learning models. The paper also highlights that this type of attack does not require any special access or permissions and can work across USB 2.0, USB 3.x, and USB-C. But the research still has a deeper issue that the USB hubs can not isolate traffic between devices properly. It is just creating opportunities for these side-channel attacks. USBSnoop ultimately shows that USB traffic patterns themselves can be a goldmine for attackers since it can unintentionally leak sensitive information.

However, "Koffi et al. [22] developed a machine learning system that uses power consumption analysis to identify malicious USB devices. USB security systems that operate at the software level fail to identify hardware-based attacks because they do not monitor hardware activities. The research suggests tracking USB power consumption irregularities as a method to detect harmful device behavior. The system utilizes real-time power readings which undergo Autoencoder feature extraction before LSTM and CNN models detect anomalies. The system achieves a near-perfect F1 score by detecting malicious devices through their unique power signatures which result from hidden activities. Physical-layer security monitoring emerges as an effective non-intrusive framework that strengthens USB security frameworks while protecting against evolving USB-based attacks.

Also, The researchers in this paper [22] examine the ability to detect suspicious device behavior conducting techniques including clustering, support vector machines (SVMs), and neural networks. The researchers also conducted a study of many machine learning algorithms for anomaly detection, focusing on their application to USB peripheral security systems. Also, Evaluation on Finally, The study demonstrates how machine learning models can be used to upgrade USB security by detecting new and evolving threats, particularly by using unsupervised approaches.

Though the enhancement of using USB peripherals has brought convenience to users, it has shown a new attack surface for hackers at the same time. The reason is that USB devices can act as dangerous tool, as they can execute harmful activity, be programmed to steal data, and inject harmful commands. To overcome this problem, researchers of this paper [17] developed a keypress attack detection system,

which uses keystroke dynamics to distinguish between legitimate typing and malicious ones. This system is trained by recording a user’s typing style with machine learning models and alerts when abnormal patterns are detected, meaning that one typing does not match the records of the system. Though the system performs very effectively within its measurements, there are some scope for improvements in it like- handling capital letters, special characters, and user mood variations. To wrap it up, with further improvements, this method can be a powerful cybersecurity tool to tackle USB keypress attacks, making our digital world a safer place.

Hernandez et al. (2017) [6] present a technical tool designed to inspect USB device firmware for hidden or potentially malicious functionalities. The main concern conveyed is- USB devices can sometimes pretend something that they are actually not like that. For example, some USB devices act as simple storage devices but they are actually capable of injecting keystrokes secretly on keyboards. To prevent this the researchers of this paper developed FirmUSB, a framework that uses a technique called symbolic execution. It allows the tool to explore different paths the program could take instead of running the firmware on the actual hardware. Symbolic execution also helps to uncover what the device might do in some certain conditions. The team customized their tools to handle that specific architecture and integrated them with well-known analysis engines like ANGR and FIE since many USB devices run on 8051 microcontrollers. The deep understanding of USB devices’ structure makes the FirmUSB special- it knows what descriptors, endpoints, and class behaviors to look for, which makes its analysis more targeted and effective. Practically, FirmUSB is capable of detecting where seemingly harmless firmware was hiding dangerous capabilities. The findings basically point to the importance of firmware level validation when it comes to trustworthy device claims but in reality it is not. This work contributes a powerful tool for the security vetting of USB firmware, offering valuable insight into the opaque operations of embedded systems.

Additionally, The authors of this paper [4] have used a broad portion of USB technology for connectivity and data transport. Bad USB threat is a source for hackers, it allows malware to be embedded in the USB framework and remain undetected by security application systems. This research aims to know the vulnerabilities associated with this device and recommend security controls that are based on the NIST Risk management framework, and categorized into Technical, operational, and other controls. They have used the NIST SP 800-53 (rev 4) category in the technical, operational, and management controls. USB connections and logging were handled by technical controls. Critical information was created by management. Furthermore, they have used DriveCom and Duck encodes for the proper utilization of communication. The study validates its effectiveness against simulated BadUSB attacks, demonstrating its potential for enhancing system security.

Qaddoori and Ali [24] introduced a compact embedded anomaly detection solution for smart meters that employs ML algorithms to monitor strange power consumption patterns. This system operates on a Raspberry Pi 4 Model B unit to perform efficient anomaly detection in energy patterns with low computational resource requirements. The system applies a two-step machine learning approach with clustering algorithms (DBSCAN and Mean-Shift) identifying daily abnormalities and

then using supervised classifiers (Decision Tree and Naïve Bayes) for detection plus unsupervised models (One-Class SVM, Isolation Forest, Elliptic Envelope) spotting periodic irregularities. The framework implements ongoing knowledge acquisition through data aggregator upgrades enabling model retraining using incoming meter measurements before updating smart meters. The system maintains effective usability through the combination of Real-time data exchange based on MQTT protocol alongside embedded security features and Bokeh web-based visualization capabilities. Experimental data reveals that DBSCAN delivers the best cluster results yet Decision Tree demonstrates perfect classification accuracy alongside One-Class SVM demonstrating the highest efficiency for periodic anomaly detection. The system is well-suited for IoT-based smart metering because of its low resource usage (29% CPU utilization with 2.31% memory use and 4.2W power draw). The proposed system delivers enhanced privacy protection through smart meter-based local operation along with faster detection speed and improved adjustment to changing power usage patterns while surpassing conventional cloud-based anomaly detection practices. The proposed framework decreases energy loss while boosting monitoring functions and provides scalable smart grid capabilities that will be improved through future work on cybersecurity developments and deep learning system integration.

The study [38] proposes a smart and practical approach to protect computers from threats carried through USB devices. In case of detecting new or unknown malware, traditional antivirus programs often fail in doing so; they are also very vulnerable to zero-day attacks. To address this issue, the researchers developed a system that checks the files against known malware databases with the help of VirusTotal API and also uses artificial intelligence to detect suspicious file behaviour. Both of these approaches combined allows the system to spot both previously seen and unseen threats. The program observes USB ports by scanning any recently connected devices and automatically takes action in real-time; it quarantines files with unusual behaviour and alerts the user via desktop pop-ups, emails or basically messages through any messaging service. The system is built with Python and machine learning libraries like Scikit-learn and TensorFlow. It is very lightweight and efficient, as it is able to scan a 16GB USB device in no more than ten seconds without straining the computer hardware. The program is also cross platform, working flawlessly on Windows, Linux and macOS, making it suitable for usage in various fields. The research emphasizes how artificial intelligence is capable of being used effectively to enhance cybersecurity in real-time, offering a necessary defense against modern USB-based threats.

Additionally, in this article [20], Fernandes and Lina explore how a small, custom-made device using an ATMEGA32U4 microcontroller and an ESP8266 WiFi module, can be turned into a wireless fake keyboard, sending keystrokes to a computer without any physical interaction. The point of this thesis is to show how attackers could use such a device to take control of a computer remotely, simply just by plugging it into a USB port and transmitting commands over WiFi. The researchers successfully built and tested the device, showing how easily this kind of attack can happen. The main purpose of the study is to raise awareness about this security risk, not promoting hacking. Besides, it also encourages to strengthen the defenses by limiting USB access to both individuals and organizations to secure their devices

more effectively.

On the other hand, Solairaj et al. (2016) [5] discusses how to detect software keyloggers- those sneaky programs can record everything you type, and can be detected before they cause harm. These keyloggers are designed to run silently and avoid detection by antivirus software. The researchers of this paper explore several smart detection strategies to detect the keyloggers. One approach looks for programs that hook into the system to watch you do. Another approach is called HoneyID uses fake input to trick the spyware and expose them. They also use a different technique called Spearman's Rank Correlation to detect bots by analyzing suspicious behaviors like- accessing files and sending keystrokes. A more advanced method they explore, called Dendritic Cell Algorithm, inspired by how the human immune system works, helps to identify abnormal behavior on a system. The paper even covers real life examples, such as- a case where an iPhone's sensors were used to capture keystrokes from a nearby keyboard. Finally, the author highlights mobile keyloggers as a growing threat and suggests using machine learning methods like Support Vector Machines for future detection since the study focuses on improving user safety and data protection.

Here, the paper [1] outlines the dangers of everyday USB devices by describing how attackers can reprogram USB devices to hide their identity. For instance, turning a flash drive into a fake keyboard that can automatically type malicious commands or hide malware in what seems like a safe storage drive. These kinds of attacks are tricky because computers typically trust USB devices the moment they are connected. The authors look at current ways to protect against these threats like blocking unknown USBs, isolating suspicious devices in secure environments, and tracking unusual behavior. But the current methods have their limits since the USB protocol is flexible and does not have the built-in security. The report ultimately asserts that we require more robust, sophisticated, and intelligent security systems capable of detecting and responding to anomalous or suspicious USB activity in real time to avert breaches of security.

Seo and Moon [7] made a detailed examination of the BadUSB vulnerability, pinpointing the proposed threats from USB devices capable of posing as imitation peripherals by substituting their firmware. The authors underlined that conventional antivirus software is unable to identify such attacks, as they normally scan USB devices' storage partitions and not the firmware level, where malicious code is present. Their approach divides the USB firmware into read-write and read-only areas, and suggests a two-phase solution to guard against it: hash-based verification of read-only descriptor fields, and digital signatures on changeable firmware code. This can catch any unauthorized modification early, and only authorized firmware updates be applied. They also had a secure update process for firmware, providing extra protection. This work formed the foundation of hardware integrity-based USB security solutions, through showing that a change of paradigm in emphasis, toward low-level device activity, could be the foundation for an effective defense.

Blanchet [8] offers a detailed exploration of the BadUSB attack, describing the covert dangers hidden in the firmware of seemingly harmless USB devices. Unlike

traditional malware, BadUSB attacks bypass typical detection by exploiting the reprogrammable area of USB microcontrollers to masquerade as trusted peripherals such as keyboards or network cards. This permits surreptitious operation of malicious commands, traffic redirection, or data leaving without raising suspicion. Blanchet calls out the big security vulnerability as the overall lack of protection at the firmware level—specifically the lack of code signing and integrity checking. Her work generalizes from USB drives to all USB-attached peripherals, highlighting the systemic defect in how USB trust is managed today. This work confirms industry’s urgent need to adopt firmware authentication protocols to block such low-level, long-term attacks.

Karystinos et al. [13] proposed Spyduino, a hardware attack proof-of-concept that leverages an Arduino UNO board that was compromised to exploit the BadUSB vulnerability to emulate a Human Interface Device (HID). Unlike the conventional USB-based malware which requires payloads or storage, Spyduino runs a pre-programmed script by emulating keystrokes and can thus carry out malicious actions like switching off firewalls, stealing user information and uploading files to an FTP server without seeking any user consent. The novelty of Spyduino, however, is its stealthiness, or the fact that it is built into a standard USB keyboard and does not need special drivers, which makes it practically undetectable by a conventional antivirus. The authors emphasize that Spyduino can be employed to attack multiple operating systems, exploits passive FTP to avoid detection and leaves minimal artifact in system logs. Such countermeasures are mentioned as device control software, behavior monitoring software such as Duckhunt, and physical port blockers. The authors note that the measures usually do not apply to stealthy hardware-based attacks.

Next, A paper [25] indicates keyloggers as a serious cybersecurity threat that is capable of stealing and manipulating sensitive information like personal data and passwords. The situation is worsening as the keyloggers are getting more sophisticated and evading traditional antivirus detection methods. The paper studies numerous detection and prevention techniques such as the use of network traffic analysis, biometric authentication, antivirus, and antimalware software to tackle this matter. To protect sensitive data, many processes are introduced including the use of advanced encryption, user education on social engineering, and conducting regular system updates. This study’s future work focuses on developing a stronger biometric system involving machine learning and artificial intelligence to enhance detection and prevention capabilities. Finally, the study emphasizes the need for collaboration among users, researchers, and industry professionals to put effort into sharing the best practices and threat intelligence to overcome this situation against keyloggers.

Similarly, This paper [18] explores serious security concerns involving stealing data and unauthorized software installations related to keyloggers. To begin with, effective monitoring tools are needed that can track user activities without being detected by any antivirus software. This paper talks about developing a software-based keylogger using Python, which records keystrokes and analyzes typing patterns to upgrade security measures. In this process, a real-time keylogger was created that logs keystrokes and clipboard data while ensuring the recorded information is safely

stored and analyzed. Also, a technique named feature selection was implemented to upgrade the accuracy and efficiency of the detection process. For future work, the authors focus on exploring keylogger's application in broader contexts involving risk assessment and behavioral analysis. They also suggest upgrading keylogger's stealth capabilities to avoid detection by antivirus software. The study aims to improve Internet users' security and privacy by providing a framework for online keylogger detection.

This study [21] also explores the growing field of a type of malware named keyloggers that poses a significant threat to personal and professional data security as it stealthily records user keystrokes. It is challenging for users to protect themselves because keyloggers can operate undetected. This study shows a strategy involving the development of a stealthy software-based keylogger using Python to monitor the activities of a victim without informing them. This keylogger has various features including capturing keystrokes, clipboard monitoring, screen capture, and microphone activation for comprehensive data collection. This process spreads the malware through external storage devices or email keeping it hidden from antivirus software. The future work of this paper focuses on improving user awareness, upgrading detection methods, and developing stronger security measures to prevent these stealthy applications. This study also focuses on the need for ongoing research into the legitimate uses of keyloggers and ethical implications such as forensic investigations and parental monitoring to balance data security with privacy concerns.

On the other hand, Denney et al. [11] have used the USB-Watch for the hardware system to detect malicious USB events in recent times. In this function the USB devices host computers the USB communication for detecting abnormal activities. It takes user input to analyze the threats through timing variations and data stream patterns. The system functions autonomously from the host operating system which results in both efficiency and lightweight performance. The accuracy of USB-Watch has been confirmed through testing which yielded a Receiver Operating Characteristic (ROC) score of 0.89 demonstrating its capability to differentiate between safe and dangerous USB activities. The research demonstrates that hardware-assisted tools represent a dependable defensive mechanism for protecting USB systems.

A paper [29] discusses three methods to detect and mitigate the BadUSB attack, which targets the USB device firmware. As the traditional antivirus is unable to detect the changes in firmware, the researchers evaluate three forensic techniques. Firstly, the signature-based detection: this method utilizes the known patterns of bad behavior like a checklist (70-80% success rate), as the hackers do not leave any optimal clues in most of the cases, so this method cannot recognize the firmware attacks. Secondly, the Behavioral Analysis: this technique notices how the USB devices behave in real-time. If the devices act differently from normal behavior like the speed of the key log or sending suspicious commands, this method is able to detect (85-92% success rate). But the flaw of this method is- it takes a long time to detect anonymous behavior. Also sometimes it makes mistakes that consider normal behaviour of the system as harmful. Finally, machine learning: by this method, computers can acknowledge the bad and good USB devices. With very few mistakes, it can detect the attacks (92-95% success) within a shorter time than the behavioral

analysis. The researchers tested these methods on different Operating systems by creating bad USB and found the third method is the best among all. The paper recommends that, as the use of USB devices is increasing day by day, using the mix of three methods for better security and points on the state where it needs to improve more for security.

Finally, In this era, threats to security emerge from USB drives because they offer quick file storage alongside unrestricted device access. Attackers use USB drives to spread malware while infecting production devices and performing data theft by accessing computers physically. An attacker can recover information from a computer's memory even when the device is completely switched off. USB drives that are lost or stolen represent a major security threat because unsecured backed-up data remains vulnerable to unauthorized access. Protect your data from threats by never using unfamiliar USB drives alongside enabling encryption, passwords and separating personal and work storage, and turning off Autorun functionality while keeping both security software and systems fully updated. This document emphasizes the role of USB devices. This document [15] was published by the CISA and they suggested reducing the critical factor vulnerabilities associated with USB device usage.

2.3 Summary of Key Findings

This study builds upon existing research and foundational studies which revealed important trends and challenges alongside methodologies for USB security. Below is a summary of the key findings:

Robust Multi-Layered Security is Crucial

The research demonstrates that USB security requires multiple detection methods because a single approach fails to protect against all USB threats. The number of USB threats continues to evolve into sophisticated and complex forms each day. A multi-layered defense strategy delivers the most effective protection because it combines hardware checks with behavior analysis and software validation. The multilevel security model improves threat recognition accuracy while reducing improper alarms and detecting new emerging risks. Modern USB security solutions depend on these security measures to function as their essential foundation [23].

USB Devices as Vectors for Sophisticated Attacks

Malicious USB devices, like ATtiny85 microcontrollers and USB Rubber Ducky, can disguise themselves as legitimate peripherals while secretly running harmful programs in the system which leads to threats. Since traditional security solutions focus on software-based threats, they fail to detect device-level vulnerabilities. Due to the limitations of traditional security measures, it is highly required a stronger security framework. Therefore, the security frameworks must incorporate hardware verification, firmware integrity checks, and behavior-based anomaly detection to prevent unauthorized USB activity and mitigate potential risks [36].

Evolving Patterns in USB Threats

Behavioral analysis techniques like keystroke pattern and power consumption anomaly

lies have proven the importance of sophisticated detection methods. Using machine learning models in recording the USB activity, the trained system can distinguish between malicious USB devices and legitimate ones. The study showed that this approach is proactive and it allows security systems to detect new attack vectors and adapt to evolving threats which make it a highly effective modern USB security solution [22].

Real-time detection Enhances System Integrity

Identifying threats as they emerge heavily relies on real-time monitoring of USB activity, along with metadata validation and anomaly detection. Studies show that real-time systems greatly reduce the response time for malicious activities, which minimizes the damage by a significant amount. That being said, we have to take into account the system optimizations, to ensure that they do not interrupt the actual USB operations [33].

Need for Scalable and Adaptive Frameworks

The usage of USB devices is widespread across various environments, from personal computing to critical infrastructure and so on and so forth. Pre-existing solutions are sometimes unscalable, and this limits their usefulness in IoT ecosystems. Thus, its scalable frameworks (that also have proper performance and accuracy) are essential for widespread adoption [37].

Balancing Security and Usability

The main lesson learned from this research shows that security protocols must not interfere with authorized USB operations. The proposed framework needs to achieve both strong protection measures and user-friendly design elements to gain successful adoption [26].

Conclusion of Key Findings The review demonstrates that an effective solution requires a security framework that combines multiple adaptive layers and user-focused protection methods. The proposed research merges advanced techniques including real-time monitoring with behavioral analysis and scalable implementation to bridge existing solution gaps and establish new USB security benchmarks.

Chapter 3

Proposed Methodology

3.1 Design Process or Methodology Overview

The design methodology of this research draws from a multi-phase model that operates to sense, avoid, and respond to BadUSB-based attacks as well as abnormal device behavior. The process is led by the core objective of developing a smart and responsive USB security system able to detect BadUSB-based attacks as well as abnormal device behavior in real-time. The process flow of this process is graphically depicted in the system architecture flowchart (Figure X), which indicates both deterministic logic as well as machine learning-based decision modules.

The procedure begins at the device connection point of entry, where a USB device is checked against a trusted whitelist. When the device is unknown or already labeled, it is blocked immediately, its attempt recorded, and the user informed, stopping any subsequent interaction. In cases of known or new but valid devices, a CAPTCHA-based verification procedure is invoked to identify human presence and avoid automatic or pre-programmed use of devices.

Following successful CAPTCHA traversal, the device is in a monitoring mode, where it performs behavioral profiling. This includes keystroke pattern monitoring, USB metadata, power consumption pattern monitoring, process execution, and communication. A machine learning classifier is now invoked to determine if device activity conforms to normal expectations. This ML-driven anomaly detection component is meant to distinguish normal human interaction from potential BadUSB usage based on learned patterns during training.

If the activity is suspicious, then detailed analysis is done using power consumption profiling by comparing the device live draw with known profiles. The system then executes a binary response: permit device access and continue monitoring, or execute the threat detection and mitigation procedures, both guided by machine learning programs trained on known bad signatures and out-of-the-box behavior.

The discovered threats are then checked for mitigation actions such as device isolation, killing processes related to it, and access restriction, and after which the device is added to the blocklist. The decision points and their outcomes are recorded for forensic traceability purposes and future model retraining.

Such a design is struck between imposing security and openness to users such that the trusted users experience minimum friction while threats are proactively intercepted. With the addition of CAPTCHA challenges, behavioral analysis, power profiling, and machine learning analysis, the system continues to be scalable in enterprise or consumer computing environments

3.2 Preliminary Design or Design (Model) Specification

The design phase of this research focused on creating a strong and scalable machine learning framework that can classify USB devices as either malicious or non-malicious. This was done using a combination of forensic information, behavioral data and device metadata. Our main goal was to build a pipeline that could identify USB-based threats while reducing false positives that might affect regular system operations. This section talks about the design principles, structure of the classification system, feature strategy and model choices before implementation.

Dataset Overview and Feature Layout

The dataset we used for model design included 2,653 records gathered from real USB usage scenarios involving both malicious and benign devices. The binary target variable referred as: 1 = Malicious device and 0 = Non-malicious device. The dataset was fairly balanced, including 1,436 non-malicious entries and 1,217 malicious entries. This balance made it suitable for supervised binary classification without the need for significant oversampling or undersampling.

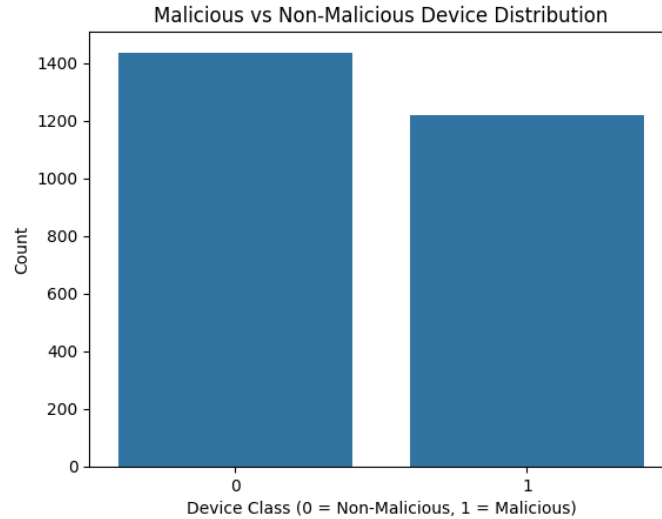


Figure 3.1: Malicious vs Non-Malicious Device Distribution

Each record included a complete set of 33 features, which can be broadly categorized as:

- **Behavioral Metrics:** power_mA, cpu_avg_percent, process_count, avg_process_runtime_sec, keystroke_mean_ms, keystroke_std_ms

- **Forensic Indicators:** ftk_available, drive_interface, removable_drive, source_type, md5_hash, image_file, sha1_hash
- **Hardware Metadata:** manufacturer, device_description, vendor_id, product_id, location_id
- **Hash Verifications and Timestamps:** verification_md5, acquisition_started, acquisition_finished

This diverse composition captures both **hardware-level traits** and **behavioral anomalies** to ensure pattern recognition.

Feature Selection and Data Reduction Strategy

To prevent overfitting and improve generalization, several domain-focused and statistical reduction techniques were used to refine the feature set:

Redundancy Elimination: Highly related features such as tracks_per_cylinder, bytes_per_sector, and sectors_per_track were combined into representative variables like sector_count.

Data Leakage Prevention: Features like serial_number, timestamp, and cryptographic hashes (md5_hash, sha1_hash) were excluded from model training to avoid leaking ground truth.

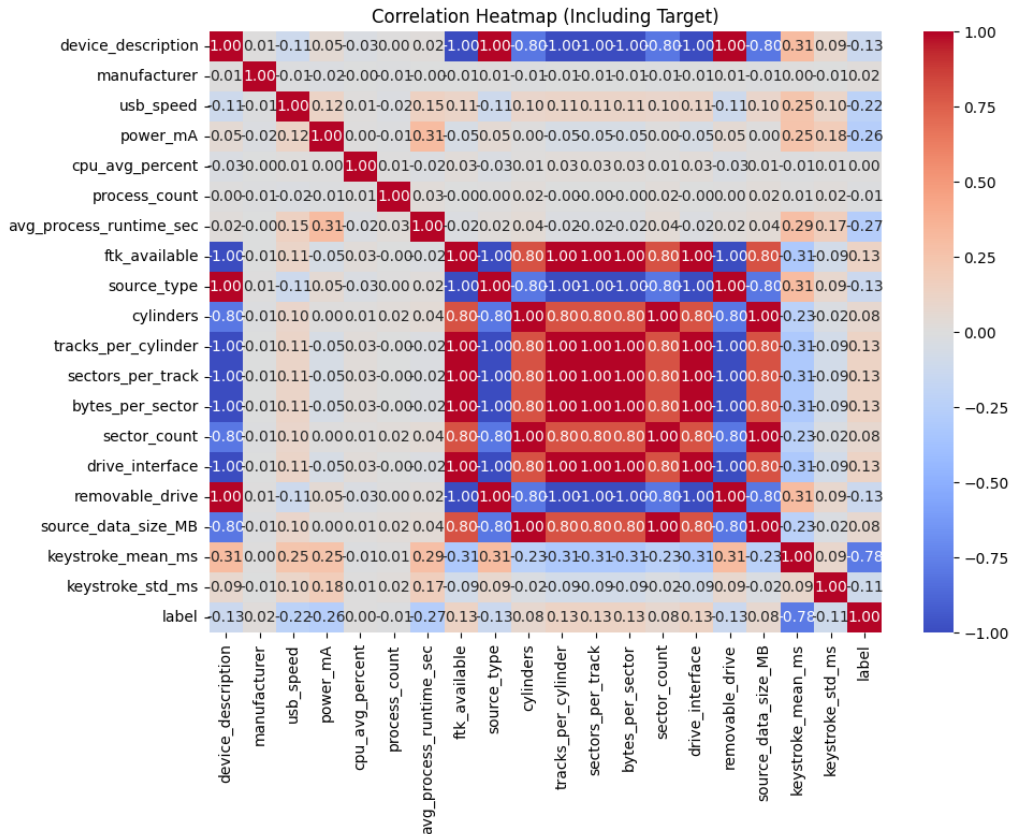


Figure 3.2: Correlation heatmap of all features including the target label

Correlation Analysis: A heatmap showed the Pearson correlations between input features and the label. Features like `source_type`, `cylinders` had redundant correlation and were removed.

Feature Importance (Model-Driven): Tree-based models like Random Forest were later used to check which features had significant predictive power. This confirmed the importance of the retained metrics such as `keystroke_mean_ms`, `power_mA`, and `avg_process_runtime_sec`.

This multi-step filtering process resulted in a clear and informative feature set that is suitable for lightweight and understandable classification.

Machine Learning Model Design

The model design focused on evaluating a variety of classification strategies to find which algorithms fit the dataset best. The following models were chosen:

Logistic Regression: A fundamental linear model that is quick and easy to grasp.

Random Forest Classifier: This model Utilizes an ensemble approach, is robust against noise, and reveals internal feature importance.

Decision Tree Classifier: It Provides straightforward, rule-based decision-making, is interpretable, and operates quickly.

K-Nearest Neighbors (KNN): This model is Based on distance metrics which is effective for identifying behavioral patterns.

Naive Bayes: A straightforward probabilistic model that performs effectively with independent features.

Each model was assessed within a consistent evaluation framework to guarantee an equitable comparison.

Classification Pipeline Architecture

The classification pipeline was designed to support modular, reproducible, and scalable experiments. It included these stages:

Preprocessing:

- Missing values were handled using `SimpleImputer` with a mean strategy.
- Categorical features like `device_description` and `manufacturer` were encoded using label encoding or one-hot encoding as needed.
- Features were standardized with `StandardScaler` to help distance-based algorithms such as KNN.

Training-Testing Split:

- Data was split with `train_test_split()` using a stratified method to maintain label distribution.
- About 70% of the data was used for training, and 30% for testing.

Model Evaluation:

- Each classifier was assessed on the same test set using key metrics:
 - **Accuracy:**(overall performance)
 - **Precision:**(correctness of malicious identification)
 - **Recall:** (sensitivity of malicious detection)
 - **ROC-AUC Score:**Score (probability-based separation)
- Confusion matrices and ROC curves were created for visual inspection.
- Bar plots were made to compare metrics across models.

Interpretability and Deployment:

- Feature importance visualizations from tree-based models helped in understanding the results.
- The design allows for easy expansion to real-time USB detection environments in future work.

3.3 Data Collection

The dataset exploited in this study was iteratively constructed to model and record realistic USB device operations in cases of security and compromised situations. It comprises 2,653 single records where each record is a single instance of USB connection. The forensic documentation was made on a controlled, Windows-based forensic system, with a combination of the forensic acquisition tools and real time monitoring steps of the system. To show as broad a variety of devices as practically common in real usage, the dataset is explicitly diversified, including common flash drives and external hard disks, as well as more advanced HID-based devices like keyboards, mice and programmable microcontrollers such as the ATtiny85 vulnerable to BadUSB-type attacks.

Each sample has 33 descriptive features which in combination create a complete picture of the identity of the device as well as the layout of its storage, the mode of interaction with the system and finally interaction patterns by the user, where relevant. The physical attributes and identification of each USB is assigned as metadata fields as `drive_model`, `drive_interface`, `removable_drive`, `device_description`, and `source_type`. In the case of storage based systems, Low level geometry including cylinders, tracks per cylinder, sectors per track and bytes per track is also stored in detail. These structural areas are important in identifying the difference between the storage devices and the in input only or the maliciously programmed devices.

Distribution of Benign vs Malicious USB Records

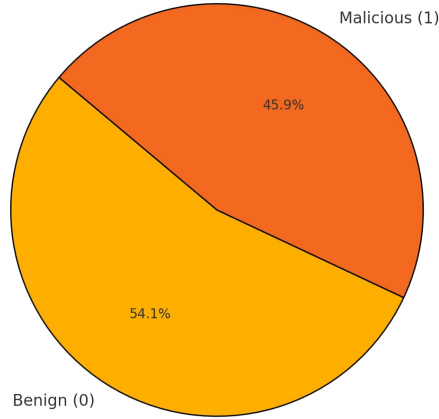


Figure 3.3: Distribution of Benign vs Malicious USB Records

Along with metadata, forensic acquisition information of the dataset consists of its `source_data_size_MB`, `image_file`, cryptographic hashes (`md5_hash`, `sha1_hash`), and acquisition start and end timestamps. The digital evidence was also traceable with the help of such tools as FTK Imager that was used to collect this information. Devices that contain no FTK reports, e.g. devices that do not exhibit mass storage (ATtiny85 operating as a keyboard), were augmented with behavioral metadata (e.g. `keystroke_dynamics`; `keystroke_mean`, `keystroke_std`) to aid differentiation of human-typed input, e.g. scripted payloads.

One of the elements of the dataset that would become critical is the label field, where each case is classified as either malicious (1) or benign (0). Such labels were given depending on the known setting and the behavior seen on each device. The missing or inapplicable values that are especially frequent in such areas as the number of sectors in non-storage devices were processed by a mark with the sign of Null, Unknown and subsequently processed as NaN when it became possible to preprocess it efficiently.

Data has been generated to facilitate machine learning of anomalous detection and identification of malignant USB behavior. It allows comparing different algorithms, including K-Nearest Neighbors (KNN), Logistic Regression, etc., and contributes to the development of the proactive and data-driven USB security systems. Broad coverage of the available range of devices, coverage of the functionality, and the good proportion between benign and harmful actions make the dataset an excellent basis to undertake specific forensic, cybersecurity, and machine learning studies in USB threat detection.

3.3.1 Data Cleaning

The dataset had some missing values, along with inconsistent formatting and irrelevant attributes. To clean our dataset, we took the following measures:

Handling missing values: Several columns contained missing values represented as string data types, such as "Null" and "Unknown". First, we have converted these into actual NaN. Several columns (cylinders, tracks_per_cylinder, sectors_per_track, bytes_per_sector, sector_count, source_data_size_MB, removable_drive) had missing numerical values, as these features only exist in USB storage devices, so we have replaced these missing values with 0.

Categorical Normalization: Some columns had missing values that were filled with 'Unknown'. We have converted them into title cases for consistency and later mapped them manually with integers.

Duplicate value removal: We have then identified all the duplicate values from our dataset and removed them.

Feature reduction: We have identified some irrelevant, redundant, unique columns (vendor_id, serial_number, location_id, product_id, alt_usb_descriptor, drive_model, md5_hash, sha1_hash, image_file, acquisition_started, acquisition_finished, verification_md5) and dropped them. We removed these features as they offered no predictive value and might have introduced noise into the learning process.

We made sure not to drop any features or attributes that could have significance in the learning process. This cleaning process ensured a compact and meaningful dataset.

3.3.2 Data Transformation

We have done some data transformations to enhance model compatibility.

- USB_speed was a text-based indicator. We have standardized it to Mbps as a numerical value. We have changed the data type from string to integer.
- We have also cleaned the string of some categorical features before label encoding. We stripped the strings and transformed them into title cases for standardization.
- We have used label encoding for categorical fields, which are device_description, source_type, drive_interface, and removable_drive.
- We have also handled some missing values in the manufacturer column by replacing them with 'Unknown'.
- After generating a heatmap from the preprocessed and encoded data, we identified the columns that had low correlation (tracks_per_cylinder, sectors_per_track, bytes_per_sector, source_type, device_description, ftk_available, cylinders, source_data_size_MB) and dropped them.
- We have normalized the numerical variables using StandardScaler to maintain equal weight across the features.

These transformations ensured that our dataset was in a fully numeric and standardized format, making it more compatible with the learning process. We converted the categorical data from text to meaningful numerical representations and normalized feature scales. We made sure that the transformed dataset preserved all the relevant attributes while minimizing bias from inconsistent data types. This preprocessing helped us achieve effective model training as well as more accurate classification performance.

3.3.3 Data Integration

Data integration in this study entailed the aggregation and combination of forensic, behavioral and system-level events on different USB devices on different platforms and tools. This was required to aggregate heterogeneous observations to a structured dataset that could be learnt by a machine to detect anomalies.

Initially, metadata and interaction logs of USB devices were collected in a methodical manner with the help of a proprietary Python-based forensic logger. Important properties that were captured by this logger included Vendor ID, Product ID, serial number, device description, USB interface speed, power consumption estimates, and active CPU usage during USB activity. It also tracked keystroke dynamics of HID-compatible devices (e.g. keyboards, ATtiny85, Raspberry Pi Pico) by computing the mean and standard deviation of keypress times. In case of no keystroke being detected within a finite time window, respective features were tagged as Null and default-labeled as non-malicious to avoid misclassification.

At the same time, the script reads FTK Imager output reports (when present) to retrieve storage-level forensic metadata such as drive geometry (cylinders, tracks, sectors), hash values (MD5, SHA1), file system layout, and acquisition timestamps. Because not every USB device presents storage volumes (e.g. ATtiny85 BadUSB), the integration logic was designed to handle the absence of FTK data by logging alternative identifiers provided by the Windows USB device manager (Get-PnpDevice) including device class, interface type, and descriptor snapshot.

To further enrich the dataset, the system gathered run time statistics like the number of active processes and the average process lifetime when using the device. These features assist in detecting possible system-level abnormalities that are brought about by malware-infected USB activity.

All collected data were dynamically appended to a centralized CSV data repository containing both benign and malicious data jumbled together. The consolidation provided homogeneous schema compatibility and preprocessing suitability by unit normalizing (e.g., time in milliseconds, storage in MB), categorical feature encoding, and unifying missing values across modalities. This aggregated dataset then provided a rich, multi-faceted picture of USB activity on 2,653 samples, which formed the foundation for the subsequent machine learning-based threat classification.

3.3.4 Data Reduction

As the dataset collected for this research included many features from USB metadata, system behavior, keystroke timings, and forensic tools, the raw feature set was both high-dimensional and varied. The data reduction process was essential to simplify the modeling pipeline, remove irrelevant or misleading fields, and improve the efficiency and reliability of the classifiers. This phase focused only on refining the input features using a combination of statistical, semantic, and empirical evaluations.

Dimensionality Reduction Strategy

The following strategy was used to perform intelligent dimensionality reduction:

1. Relevant and Context-Independent Fields Removal

Some fields, while descriptive, were found to be irrelevant for classification and were excluded:

- **timestamp:** A time-related value that does not affect device behavior.
- **serial_number, location_id:** Unique identifiers for devices that do not apply to general use.
- **md5_hash sha1_hash, image_file, verification_md5:** Static forensic values that only help with file integrity and not with predictive modeling.

These features often contain information specific to a device or session, which could bias the model or lead to data leakage, as they may uniquely identify a sample instead of generalizing malicious traits.

2. Redundancy and Multicollinearity Pruning

Storage-related descriptors like:

- `cylinders`
- `tracks_per_cylinder`
- `sectors_per_track`
- `bytes_per_sector`

showed high multicollinearity and were effectively represented by `sector_count`, which combines multiple characteristics into one value. Keeping only `sector_count` helped us avoid redundant signals without losing important information.

3. Skewness-Based Statistical Review

Using the `.skew()` function on the numerical columns, we examined the skewness of the dataset to understand how the data is distributed. Most features showed low skewness, suggesting they followed normal distributions. Features like `power_mA`, `avg_process_runtime_sec`, and `label` had mild right skew, while `ftk_available` and `cpu_avg_percent` showed slight left skew. These levels of skew were acceptable for binary classification. No transformation was needed, but the review indicated that features such as `power_mA` and `cpu_avg_percent` had significant variance, supporting their inclusion in the final model.

4. Model-Driven Feature Importance

Random Forest and Decision Tree models were used to evaluate feature importance during the exploratory phase. The top contributors included:

- `keystroke_mean_ms`
- `avg_process_runtime_sec`
- `power_mA`
- `sector_count`
- `cpu_avg_percent`
- `process_count`
- `usb_speed`
- `keystroke_std_ms`

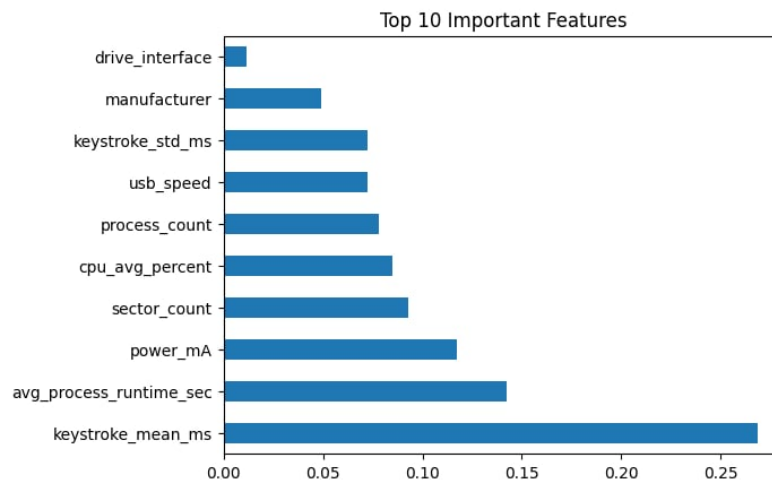


Figure 3.4: Top 10 Important Features

These features were kept, even if their variance or correlation was moderate, because they had proven to help improve classification accuracy.

Impact and Advantages of Reduction

The data reduction process resulted in several clear benefits:

- **Reduced Dimensionality (by ~60%):** This decreased the number of features from over 30 to about 12 essential ones.
- **Improved Model Performance:** By concentrating on the most useful variables, we significantly increased model accuracy and AUC across classifiers.
- **Faster Training and Evaluation:** There was less memory usage and quicker iteration time during training.
- **Lower Overfitting Risk:** We removed noisy identifiers and highly correlated fields.

- **Higher Interpretability:** Simpler models are easier to understand for practical use in cybersecurity.

Here is the heatmap after dropping unnecessary columns:

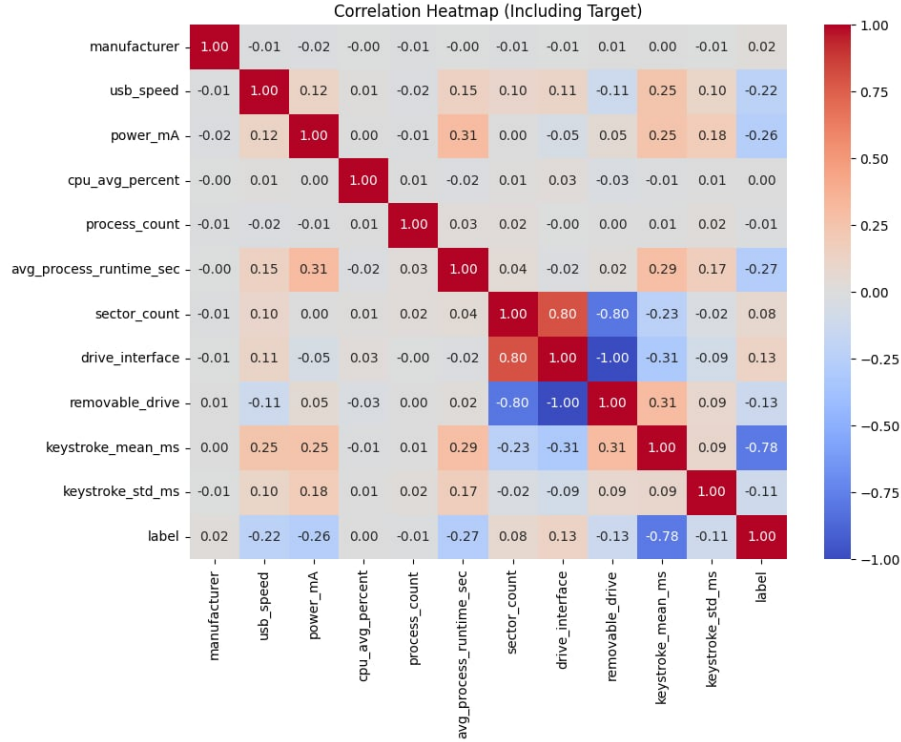


Figure 3.5: Correlation Heatmap (Including Target)

3.3.5 Summary of Preprocessed Data

The final dataset used for the design and analysis of machine learning models underwent in-depth preprocessing to ensure data integrity and consistency. The dataset contained 'Null' and 'Unknown' values in string format, so during import, these were converted into NaN values. We then dropped irrelevant, redundant, and unique columns (vendor_id, serial_number, location_id, product_id, alt_usb_descriptor, drive_model, md5_hash, sha1_hash, image_file, acquisition_started, acquisition_finished, verification_md5) to avoid noise in the analysis. Several columns (cylinders, tracks_per_cylinder, sectors_per_track, bytes_per_sector, sector_count, source_data_size_MB, removable_drive) had missing numerical values, as these features only exist in USB storage devices. We replaced these missing values with 0. The USB speed column was in string format, so we converted it into numerical form and standardized it to Mbps for consistency. Label encoding was applied to categorical features (manufacturer, device_description, source_type, drive_interface, removable_drive) to transform them into numerical features. We then generated a heatmap and dropped the columns (tracks_per_cylinder, sectors_per_track, bytes_per_sector, source_type, device_description, ftk_available, cylinders, source_data_size_MB) that showed low correlation with the target feature. To address class imbalance, SMOTE was applied to the training set after splitting. Finally, StandardScaler was used to standardize input distributions. The resulting

preprocessed dataset is complete, clean, well-structured, and numerically encoded with balanced classes, making it suitable for machine learning applications.

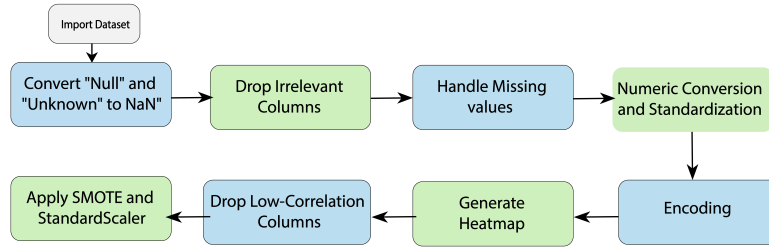


Figure 3.6: Data preprocessing pipeline

3.4 Implementation of Selected Design

Building on the architectural plan detailed in the preliminary design, the proposed system was brought to life by combining preprocessing, dimensionality reduction, and supervised machine learning algorithms into a single classification workflow. This section explains the technical execution of the chosen pipeline, highlighting how the design was implemented and improved for detecting malicious USB devices.

Development Environment and Tools:

The system was developed and tested using:

- Google Colab which is a cloud-based Python execution using web browser for interactive coding.
- Python Libraries such as scikit-learn, pandas, seaborn, numpy and matplotlib.
- Dataset, a custom-built forensic dataset `final_usb_dataset_v5.csv` that contains metadata, behavioral stats, and keystroke features for USB activity.

Execution of Data Pipeline:

This phase carries out the outlined steps of the design phase:

Missing Value Handling: We first replaced detected missing values labeled as 'Null' or 'Unknown,' then filled them in using SimpleImputer with a mean strategy for numerical fields.

One-Hot Encoding and Binarization:

- We transformed high-cardinality categorical features, such as `device_description`, using `get_dummies()`.
- We explicitly encoded binary fields, like `removable_drive` and `ftk_available`.
- We encoded the `label` column as 0 for non-malicious and 1 for malicious.

Feature Scaling For models such as KNN and Logistic Regression, we used `StandardScaler` to normalize the features. This ensures consistent distance and gradient calculations.

Dimensional Optimization and Training:

Redundant Features Dropped: We looked at the correlation matrix and Columns with high collinearity or low contribution were excluded.

Train-Test Splitting: The data was split into 70% training and 30% testing. We kept the label distribution by using `stratify = y`.

Model Training:

- We implemented models such as Logistic Regression, KNN, Random Forest, Decision Tree, and Naive Bayes.
- Each model was trained using `fit()` on the scaled training dataset.

Evaluation Workflow Implementation:

For each model, the following steps were executed:

- Predictions (`predict()` and `predict_proba()`).
- Metric Calculation:
 - Accuracy
 - Precision
 - Recall
- AUC-ROC Score (based on class 1 probabilities)
- **Confusion Matrices:** We plotted these using `seaborn.heatmap()`.
- **ROC Curve Plotting:** We plotted the ROC curves of all models together for visual comparison.

These metrics were calculated in an organized loop using lists and dictionaries to make scoring and visualization easier.

Result Visualization:

To help with comparison, we created dynamic visualizations:

- **Bar Plots:** These show accuracy, precision, recall, and AUC scores.
- **ROC Curves:** We overlaid plots to visually assess class separability.
- **Confusion Matrices:** These helped evaluate false positives and negatives, which are important for security detection.

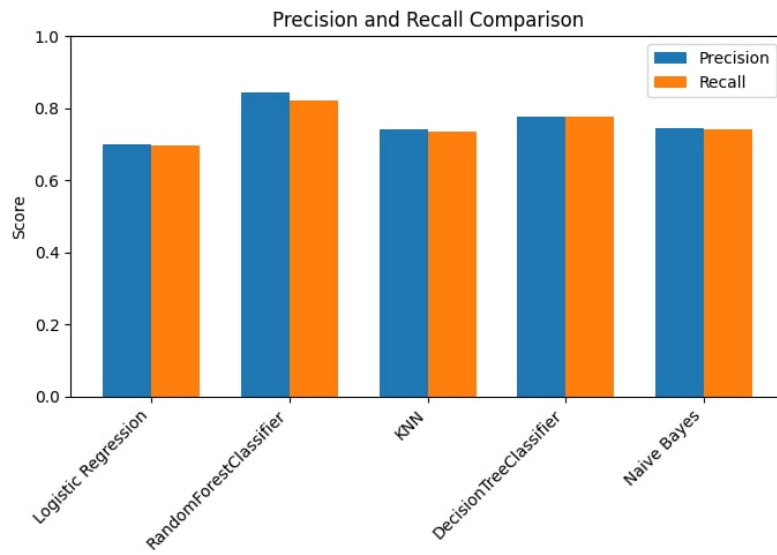


Figure 3.7: Precision and Recall Comparison)

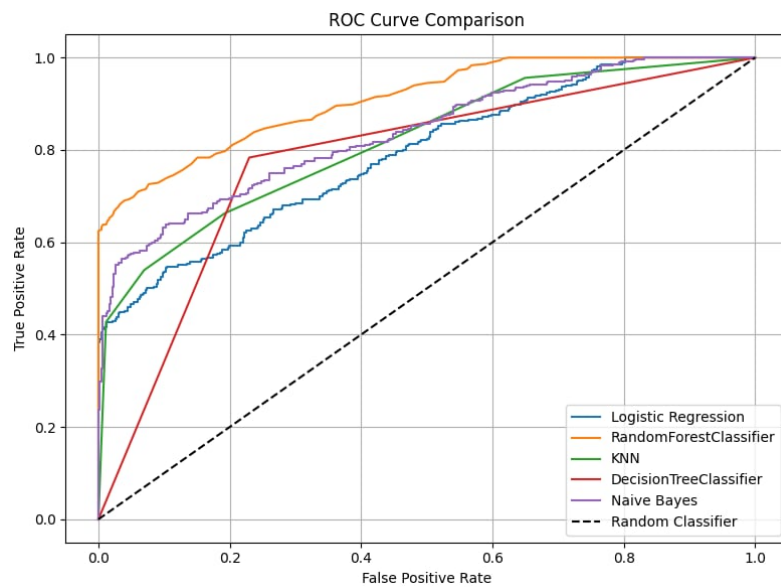


Figure 3.8: ROC Curve Comparison

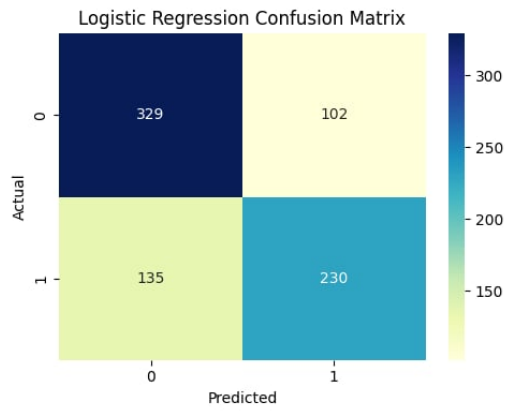


Figure 3.9: Logistic Regression Confusion Matrix

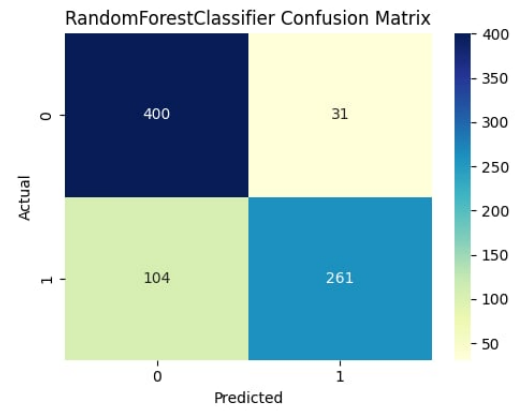


Figure 3.10: Random Forest Confusion Matrix

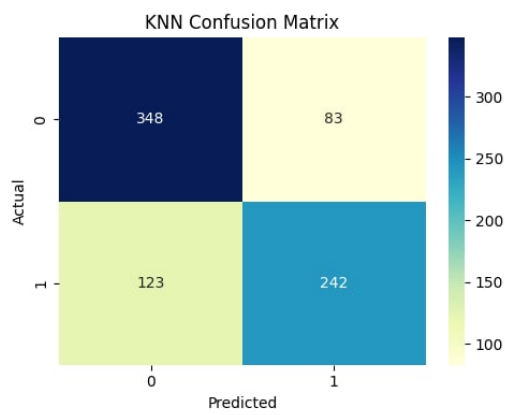


Figure 3.11: KNN Confusion Matrix

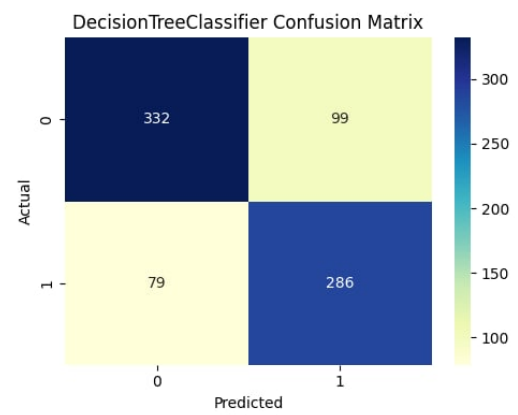


Figure 3.12: Naive Bayes Confusion Matrix

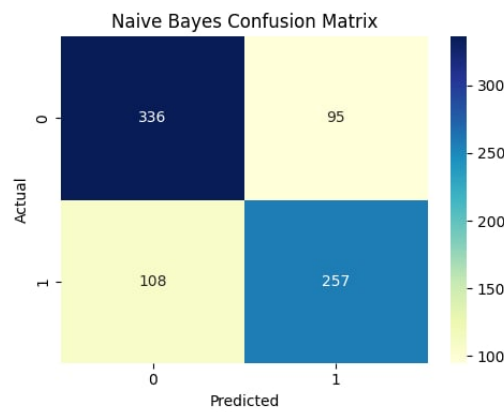


Figure 3.13: Decision Tree Confusion Matrix

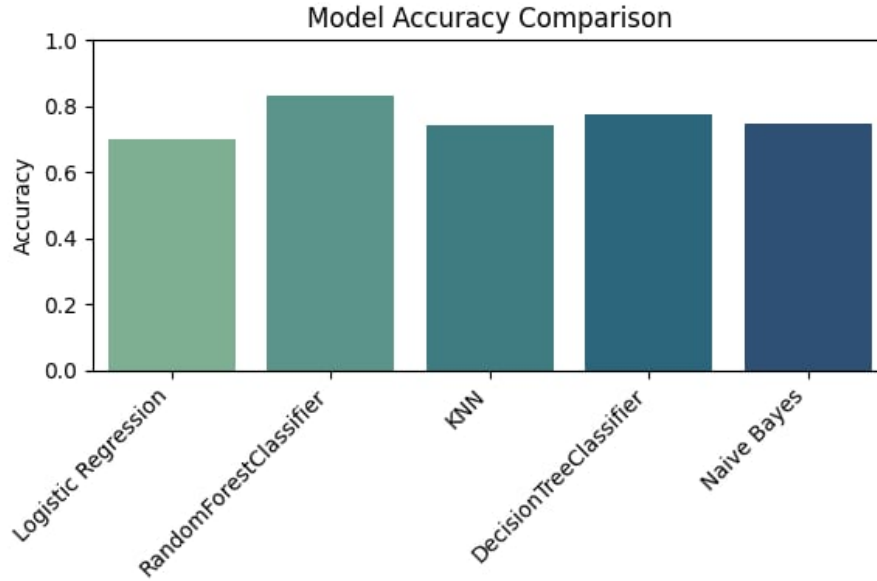


Figure 3.14: Model Accuracy Comparison

3.5 Conclusion

In conclusion, this study thereby advances the domain of USB security by contributing a practical, scalable, proactive defense system that can mitigate advanced USB-based threats. It works on a presently acute security gap and sets the stage for future endeavors in hardware-assisted cybersecurity, behavioral threat modeling, and real-time anomaly detection systems.

The ubiquity of USB use and their exploitation across personal, corporate, and critical infrastructure domains has spread the vulnerabilities, hence turning them into a central vector for fairly sophisticated cyber threats, BadUSB being an example. This thesis tackled these issues by moving beyond conventional software-only methods to design, implement, and deliver a comprehensive and intelligent USB threat detection framework. This system employs behavioral analysis, USB metadata validation, and power consumption profiling coupled with machine learning anomaly detection on the other end, to offer a fast, resilient, and adaptive security architecture.

Any traditional methods are often powerless against firmware threats or other hardware-level attacks at the darker side of the spectrum, while this layered structure sets apart legitimate USB activity from a malignant one based on finer behavioral and forensic clues; thus, keystroke dynamics and CAPTCHA checking catch nonhuman interactions, while power profiling and process-interval analysis reveal the secret malicious operations. The presence of machine learners improves the identification system's potential to generalize across multiple device types and circumstances of use, and so allows for a dynamic adaptation to changes in attack methodology while not compromising the user's experience or underway.

The entire framework was taken through the four phases of data collection, pre-

processing, and dimensionality reduction, model training, and model evaluation to demonstrate the viability and efficiency of the robust USB security framework. The experimental results have clearly shown that the framework can detect threats with a high level of accuracy and that it produces very few false-positive outputs, making it very useful for deployment in real-life scenarios such as air-gapped environments or security-sensitive infrastructure settings. Also, the modular and scalable nature of the framework allows for further developments where it could be used directly as part of a large cybersecurity framework.

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